

# Multiple comparisons

Project: Reproducible AI Research Study

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Statistical tests and assumptions

Without opening the old lesson, explain one key idea from Statistical tests and assumptions. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Uncertainty intervals

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Experimental units and independence

Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace the worked case for Multiple comparisons. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Multiple comparisons. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Apply a new case independently

Four tests have p-values .01, .02, .2, .9. Compute threshold and decisions.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

### My reasoning and evidence

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## 6. W2 Trace the worked case

Five tests have p-values .009, .012, .3, .4, .8. Show the intermediate steps and give the conclusion with its scope.

### My reasoning and evidence

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## 7. W3 Audit the limit

A report omits this qualification: Adjustment cannot repair invalid tests, post-hoc outcome selection or unreported analyses. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

### My reasoning and evidence

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## 8. W4 Apply a new case independently

Four tests have p-values .01, .02, .2, .9. Compute threshold and decisions.

### My reasoning and evidence

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## 9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

### My reasoning and evidence

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## 10. Project checkpoint and reflection

Add a labelled multiple comparisons evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of multiple comparisons. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

### My project change, supporting evidence and remaining limit

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## 11. Feedback, return check and learning record

## Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

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**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 12. Pictorial case practice: Multiple comparisons

Six declared prompt comparisons. A team performs six valid hypothesis tests and applies a predeclared Bonferroni familywise threshold.

Familywise  $\alpha=0.05$ . Supplied unadjusted p-values are 0.004,0.012,0.02,0.08,0.4,0.9. These values are stipulated, not calculated from invented data.

## Supplied p-values and six-test decisions

|        | Supplied p | At cutoff 0.00833 |
|--------|------------|-------------------|
| Test 1 | 0.004      | Reject            |
| Test 2 | 0.012      | Do not reject     |
| Test 3 | 0.02       | Do not reject     |
| Test 4 | 0.08       | Do not reject     |
| Test 5 | 0.4        | Do not reject     |
| Test 6 | 0.9        | Do not reject     |

Read a cell using both its row and column labels.

## Multiple comparisons

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Count the planned tests

There are six tests in the declared family.



### 2. Compute corrected threshold

$0.05/6$  is about 0.00833 for each test.



### 3. Apply all six decisions

Only 0.004 is below 0.00833; the other five do not reject under this rule.

### CHANGE ONE THING

The declared family has ten tests instead of six; keep the listed values unchanged.

Explain the changed result using the matching lesson.

Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.